

COMPUTATIONAL APPENDIX TO NEURAL BELLMAN OPERATORS: R29

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1. COMPLETE FIXED-COHORT RECORD

Every candidate appears below. N denotes width/depth/seed minus 27000; P denotes polynomial degree; R denotes random seed minus 29000. Raw regret is computed only after the candidate has been frozen. Raw and completed bounds are reference-free exact residual certificates. The complete statewise values, action tables, restart losses, and correction topology are retained as machine-readable files.

TABLE I: All 115 candidates at .01: raw performance, completion bounds, and action changes

d/L	Policy	Raw regret	Raw bound	Fixed bound	Budget bound	Fixed edits	Budget edits
2/4	N16/1/301	0	0	0	0	0	0
2/4	N16/1/302	0	0	0	0	0	0
2/4	N16/2/301	0	0	0	0	0	0
2/4	N16/2/302	0	0	0	0	0	0
2/4	N32/1/301	0	0	0	0	0	0
2/4	N32/1/302	0	0	0	0	0	0
2/4	N32/2/301	0	0	0	0	0	0
2/4	N32/2/302	0	0	0	0	0	0
2/4	P2	0.1391	0.1571	2.089×10^{-4}	2.089×10^{-4}	7	7
2/4	P3	0.1391	0.1571	0	0	7	7
2/4	R1	4.767	7.419	0	0	84	84
2/4	R2	5.054	8.931	2.089×10^{-4}	2.089×10^{-4}	84	84
2/4	R3	4.598	7.377	0	0	76	76
2/4	R4	4.214	7.046	0	0	75	75
2/4	R5	4.083	6.029	0	0	76	76
2/4	R6	4.134	8.743	0	0	78	78
2/4	R7	4.899	8.06	0	0	71	71
2/4	R8	3.964	6.25	0	0	73	73
2/4	R9	5.019	7.77	2.089×10^{-4}	2.089×10^{-4}	76	76
2/4	R10	4.293	6.747	0	0	76	76
2/4	Myopic	3.849	4.472	0	0	98	98
2/4	Base stock	1.935	2.017	2.089×10^{-4}	2.089×10^{-4}	58	58
2/4	Two-period	0.8426	1.111	2.089×10^{-4}	2.089×10^{-4}	32	32
3/4	N16/1/301	0.01418	0.01418	0	0	3	3
3/4	N16/1/302	0.01418	0.01418	0	0	3	3
3/4	N16/2/301	0	0	0	0	0	0
3/4	N16/2/302	0	0	0	0	0	0
3/4	N32/1/301	0.01355	0.01355	0	0	2	2
3/4	N32/1/302	0.01418	0.01418	0	0	1	1
3/4	N32/2/301	0.01355	0.01355	0	0	1	1
3/4	N32/2/302	0.01418	0.01418	0	0	3	3
3/4	P2	0	0	0	0	0	0
3/4	P3	0	0	0	0	0	0

d/L	Policy	Raw regret	Raw bound	Fixed bound	Budget bound	Fixed edits	Budget edits
3/4	R1	5.048	9.497	0	0	350	350
3/4	R2	4.797	9.901	0	0.006601	351	350
3/4	R3	4.472	8.613	0	0	344	344
3/4	R4	4.523	8.488	0	0	351	351
3/4	R5	4.496	9.133	0	0.006601	337	336
3/4	R6	5.141	11.62	0	0.006601	362	361
3/4	R7	4.578	10.21	0	0.006601	342	341
3/4	R8	4.761	8.865	0	0	351	351
3/4	R9	4.781	7.755	0	0.006601	348	347
3/4	R10	4.987	10.16	0	0	343	343
3/4	Myopic	3.59	5.165	0	0	443	443
3/4	Base stock	1.339	1.374	0	0	199	199
3/4	Two-period	0.7878	0.8156	0	0	93	93
4/4	N16/1/301	0.3666	0.4915	0	0.009483	61	47
4/4	N16/1/302	0.3666	0.4914	0	0.006551	59	46
4/4	N16/2/301	0.3666	0.4915	0	0.009483	55	43
4/4	N16/2/302	0.3666	0.4915	0	0.009483	56	46
4/4	N32/1/301	0.3666	0.4914	0	0.006551	61	48
4/4	N32/1/302	0.3666	0.4914	0	0.006551	59	47
4/4	N32/2/301	0.3666	0.4915	0	0.009483	61	48
4/4	N32/2/302	0.3666	0.4915	0	0.009483	58	44
4/4	P2	0.03934	0.03934	0	0.006551	17	4
4/4	P3	0.3666	0.4914	0	0.006551	59	46
4/4	R1	5.736	15.08	0	0.009483	1515	1510
4/4	R2	5.253	12.99	0	0.009483	1516	1509
4/4	R3	4.998	11.95	0	0.006551	1493	1491
4/4	R4	4.437	10.65	0	0.009483	1512	1506
4/4	R5	4.593	12.53	0	0.009483	1471	1465
4/4	R6	4.969	12.96	0	0.006551	1494	1490
4/4	R7	5.271	14.35	0	0.006551	1502	1496
4/4	R8	4.816	11.17	0	0.006551	1460	1456
4/4	R9	5.539	15.8	0	0.009483	1484	1480
4/4	R10	5.266	13.27	0	0.006551	1505	1500
4/4	Myopic	3.273	5.858	0	0	1763	1763
4/4	Base stock	1.245	1.29	0	0.006551	737	725
4/4	Two-period	0.8514	0.8695	0	0.002989	357	356
4/4	N32/2/303	0.3666	0.4915	0	0.009483	61	45
4/4	N32/2/304	0.3666	0.4915	0	0.009483	58	45
4/4	N32/2/305	0.3666	0.4915	0	0.009483	58	46
4/4	N32/2/306	0.3666	0.4915	0	0.009483	58	45
4/4	N32/2/307	0.3666	0.4915	0	0.009483	59	47
4/4	N32/2/308	0.3666	0.4915	0	0.009483	57	47
4/4	N32/2/309	0.3666	0.4915	0	0.009483	56	46
4/4	N32/2/310	0.3666	0.4915	0	0.009483	60	45
4/5	N32/2/301	0.3664	0.4913	3.962×10^{-4}	0.006804	112	87
4/5	N32/2/302	0.3664	0.4913	3.962×10^{-4}	0.006804	120	92
4/5	P2	0.1925	0.2019	3.962×10^{-4}	0.006804	92	62
4/5	P3	0.1029	0.1108	3.962×10^{-4}	0.006551	84	48
4/5	R1	5.827	15.86	1.856×10^{-4}	0.009483	3878	3865
4/5	R2	6.422	14.98	3.962×10^{-4}	0.006551	3867	3855
4/5	R3	6.098	13.37	3.962×10^{-4}	0.009483	3907	3896
4/5	R4	6.299	15.81	3.962×10^{-4}	0.006804	3910	3899
4/5	R5	5.839	14.05	0	0.009483	3877	3864
4/5	R6	6.337	19.6	3.962×10^{-4}	0.009483	3903	3893
4/5	R7	6.933	17.09	3.962×10^{-4}	0.009483	3901	3885
4/5	R8	5.76	12.84	3.962×10^{-4}	0.006551	3895	3885

d/L	Policy	Raw regret	Raw bound	Fixed bound	Budget bound	Fixed edits	Budget edits
4/5	R9	5.84	12.78	3.962×10^{-4}	0.006551	3883	3873
4/5	R10	5.867	13.72	3.962×10^{-4}	0.006551	3850	3837
4/5	Myopic	4.057	7.221	0	0	4183	4183
4/5	Base stock	1.884	2.231	0	0.006551	1961	1949
4/5	Two-period	1.491	1.563	1.856×10^{-4}	1.856×10^{-4}	1276	1276
4/5	N32/2/303	0.3664	0.4913	3.962×10^{-4}	0.009483	113	82
4/5	N32/2/304	0.3664	0.4913	3.962×10^{-4}	0.006804	112	82
4/5	N32/2/305	0.3664	0.4913	3.962×10^{-4}	0.006804	107	79
5/4	N32/2/303	0.2399	0.2399	0	0	24	24
5/4	P2	0.2399	0.24	0	0.006237	38	33
5/4	R1	7.066	21.65	0	0.009065	6201	6187
5/4	Myopic	3.552	6.384	0	0	7076	7076
5/4	Base stock	2.544	5.244	0	0	2485	2485
5/4	Two-period	2.72	5.302	0	0	1448	1448
6/4	N32/2/303	0.8042	0.9011	0	0	274	274
6/4	P2	0.865	0.9819	0	0	329	329
6/4	R1	7.541	25.26	0	0	25437	25437
6/4	Myopic	3.543	6.752	5.167×10^{-4}	5.167×10^{-4}	28462	28462
6/4	Base stock	4.752	11.09	0	0	8321	8321
6/4	Two-period	5.346	11.56	0	0	4600	4600
7/4	N32/2/303	1.549	2.148	0	0.009704	655	634
7/4	P2	1.809	2.245	0	0.009704	1608	1587
7/4	R1	10.41	34.27	0	0.009704	102839	102836
7/4	Myopic	3.659	7.056	0	0.004821	114348	114341
7/4	Base stock	7.072	17.73	0	0.009704	29044	29023
7/4	Two-period	8.011	18.54	0	0	13713	13713

2. STAGE-BY-STAGE COST LEDGER

An asterisk marks inherited-policy replay: its R28 fitting work is not repeated or included in the R29 generation entry. The component sum charges the full reference and model per policy and includes the raw audit and selected budget repair, not all sensitivity variants. It excludes startup and serialization, which remain in the cohort resource record. CPU and wall time are different measurements. Full model-construction, reference, policy compilation, comparison, and amortization fields remain in each JSON record.

TABLE II: Seconds by candidate; totals are non-amortized measured component sums

d/L	Policy	Generate	Raw audit	Repair	Verify	Wall sum	CPU sum
2/4	N16/1/301	$2.003 \times 10^{-4}*$	0.001766	0.001802	0.001595	0.01101	0.01101
2/4	N16/1/302	$1.801 \times 10^{-4}*$	0.001609	0.001757	0.001577	0.01067	0.01067
2/4	N16/2/301	$1.587 \times 10^{-4}*$	0.001615	0.001792	0.00159	0.01074	0.01074
2/4	N16/2/302	$1.723 \times 10^{-4}*$	0.00162	0.001765	0.001631	0.01075	0.01075
2/4	N32/1/301	$2.06 \times 10^{-4}*$	0.001635	0.001788	0.001598	0.01081	0.01081
2/4	N32/1/302	$1.765 \times 10^{-4}*$	0.001639	0.001774	0.001592	0.01072	0.01072
2/4	N32/2/301	$1.777 \times 10^{-4}*$	0.001622	0.001769	0.001587	0.01073	0.01073
2/4	N32/2/302	$2.057 \times 10^{-4}*$	0.001628	0.001781	0.001592	0.01077	0.01077
2/4	P2	$2.055 \times 10^{-4}*$	0.001714	0.001912	0.00165	0.01131	0.01131
2/4	P3	$1.771 \times 10^{-4}*$	0.00169	0.001831	0.001599	0.011	0.01101
2/4	R1	0.001038	0.001753	0.002347	0.001622	0.01252	0.01252
2/4	R2	9.675×10^{-4}	0.001756	0.002434	0.001661	0.01265	0.01265
2/4	R3	9.7×10^{-4}	0.001757	0.002277	0.001661	0.0124	0.01239

d/L	Policy	Generate	Raw audit	Repair	Verify	Wall sum	CPU sum
2/4	R4	9.524×10^{-4}	0.001758	0.002275	0.001611	0.01236	0.01236
2/4	R5	9.437×10^{-4}	0.001762	0.002582	0.001607	0.01266	0.01266
2/4	R6	9.45×10^{-4}	0.001762	0.002296	0.001625	0.01238	0.01238
2/4	R7	9.332×10^{-4}	0.001748	0.002278	0.001621	0.01236	0.01236
2/4	R8	0.001315	0.001766	0.002268	0.001601	0.01266	0.01267
2/4	R9	9.197×10^{-4}	0.001753	0.002381	0.00167	0.01255	0.01255
2/4	R10	9.247×10^{-4}	0.001765	0.002298	0.00161	0.01237	0.01237
2/4	Myopic	1.379×10^{-4}	0.001764	0.002414	0.001619	0.01172	0.01172
2/4	Base stock	1.102×10^{-4}	0.00174	0.002281	0.001659	0.01164	0.01164
2/4	Two-period	0.001391	0.001695	0.002075	0.001661	0.01261	0.01261
3/4	N16/1/301	$1.802 \times 10^{-4*}$	0.006947	0.006917	0.006565	0.04626	0.04626
3/4	N16/1/302	$1.965 \times 10^{-4*}$	0.006898	0.006945	0.006553	0.04606	0.04605
3/4	N16/2/301	$2.246 \times 10^{-4*}$	0.006596	0.006839	0.006633	0.04487	0.04488
3/4	N16/2/302	$2.454 \times 10^{-4*}$	0.006564	0.006857	0.006572	0.04459	0.04459
3/4	N32/1/301	$2.14 \times 10^{-4*}$	0.006878	0.007034	0.006655	0.04562	0.04562
3/4	N32/1/302	$2.185 \times 10^{-4*}$	0.007027	0.006949	0.006585	0.04559	0.0456
3/4	N32/2/301	$2.269 \times 10^{-4*}$	0.006908	0.006873	0.006613	0.04508	0.04508
3/4	N32/2/302	$2.173 \times 10^{-4*}$	0.006891	0.007057	0.00659	0.04568	0.04569
3/4	P2	$2.094 \times 10^{-4*}$	0.006564	0.006926	0.006578	0.0444	0.0444
3/4	P3	$2.051 \times 10^{-4*}$	0.006563	0.006873	0.0066	0.04444	0.04444
3/4	R1	0.005023	0.007098	0.009017	0.006628	0.05295	0.05295
3/4	R2	0.003786	0.00717	0.009542	0.006903	0.05299	0.05299
3/4	R3	0.003681	0.007228	0.009065	0.006584	0.05199	0.05199
3/4	R4	0.003651	0.007186	0.009127	0.006589	0.05181	0.05182
3/4	R5	0.003627	0.007222	0.009531	0.007022	0.05325	0.05325
3/4	R6	0.003655	0.007226	0.009722	0.006994	0.05353	0.05353
3/4	R7	0.003656	0.007214	0.009573	0.007014	0.05303	0.05303
3/4	R8	0.003547	0.007185	0.01152	0.006827	0.05411	0.05411
3/4	R9	0.003593	0.007149	0.009597	0.006941	0.05281	0.0528
3/4	R10	0.00369	0.007123	0.009283	0.00674	0.05193	0.05193
3/4	Myopic	4.535×10^{-4}	0.007152	0.009774	0.006689	0.04906	0.04907
3/4	Base stock	3.048×10^{-4}	0.007136	0.008217	0.006709	0.04739	0.04738
3/4	Two-period	0.005767	0.007036	0.00757	0.006601	0.05188	0.05189
4/4	N16/1/301	$3.17 \times 10^{-4*}$	0.03234	0.03227	0.03212	0.2151	0.2151
4/4	N16/1/302	$4.228 \times 10^{-4*}$	0.03262	0.03276	0.03244	0.2151	0.2151
4/4	N16/2/301	$3.897 \times 10^{-4*}$	0.03269	0.0326	0.03448	0.2176	0.2176
4/4	N16/2/302	$3.767 \times 10^{-4*}$	0.03321	0.03212	0.03251	0.2154	0.2153
4/4	N32/1/301	$3.823 \times 10^{-4*}$	0.03202	0.03207	0.03229	0.2154	0.2155
4/4	N32/1/302	$3.996 \times 10^{-4*}$	0.03195	0.03203	0.03287	0.2174	0.2174
4/4	N32/2/301	$3.904 \times 10^{-4*}$	0.03256	0.03286	0.03248	0.2162	0.2162
4/4	N32/2/302	$4.133 \times 10^{-4*}$	0.03241	0.03224	0.03235	0.2153	0.2153
4/4	P2	$3.62 \times 10^{-4*}$	0.03227	0.03183	0.03223	0.2144	0.2144
4/4	P3	$3.901 \times 10^{-4*}$	0.03235	0.03232	0.03208	0.2151	0.215
4/4	R1	0.01417	0.03351	0.04214	0.03317	0.2432	0.2432
4/4	R2	0.01416	0.03319	0.04088	0.03237	0.2386	0.2386
4/4	R3	0.01403	0.03343	0.04042	0.03232	0.2378	0.2378
4/4	R4	0.01429	0.03423	0.04139	0.03257	0.2418	0.2418
4/4	R5	0.01401	0.03407	0.04163	0.03272	0.2399	0.2399
4/4	R6	0.01406	0.03378	0.04128	0.03237	0.2397	0.2397
4/4	R7	0.01402	0.03434	0.0406	0.03235	0.2394	0.2394
4/4	R8	0.01433	0.03366	0.04032	0.03193	0.2376	0.2375
4/4	R9	0.01427	0.03376	0.04192	0.03186	0.2408	0.2408
4/4	R10	0.01425	0.03361	0.04061	0.03255	0.2392	0.2392
4/4	Myopic	0.001593	0.03313	0.04022	0.03134	0.2215	0.2215
4/4	Base stock	8.257×10^{-4}	0.03338	0.03699	0.03288	0.2221	0.2221
4/4	Two-period	0.02705	0.0323	0.03238	0.03106	0.2394	0.2394

d/L	Policy	Generate	Raw audit	Repair	Verify	Wall sum	CPU sum
4/4	N32/2/303	2.296	0.0326	0.03257	0.03271	2.511	2.511
4/4	N32/2/304	2.287	0.03332	0.03274	0.03235	2.504	2.504
4/4	N32/2/305	2.287	0.03347	0.03296	0.03241	2.504	2.504
4/4	N32/2/306	2.291	0.03377	0.03332	0.03318	2.508	2.508
4/4	N32/2/307	2.29	0.03319	0.0328	0.03296	2.506	2.506
4/4	N32/2/308	2.292	0.03383	0.03309	0.03268	2.508	2.508
4/4	N32/2/309	2.288	0.03354	0.0328	0.03278	2.504	2.504
4/4	N32/2/310	2.293	0.03466	0.03394	0.03386	2.513	2.513
4/5	N32/2/301	6.162×10^{-4} *	0.08153	0.07898	0.0785	0.5774	0.5404
4/5	N32/2/302	5.888×10^{-4} *	0.08061	0.08003	0.08133	0.5788	0.5418
4/5	P2	6.194×10^{-4} *	0.0817	0.08207	0.082	0.5833	0.5463
4/5	P3	6.596×10^{-4} *	0.0832	0.08328	0.08367	0.5881	0.5512
4/5	R1	0.035	0.08778	0.1058	0.08232	0.6499	0.6129
4/5	R2	0.03424	0.0876	0.1112	0.08576	0.6608	0.6237
4/5	R3	0.03448	0.08777	0.1084	0.08241	0.6519	0.6149
4/5	R4	0.03464	0.08494	0.1072	0.08275	0.6506	0.6137
4/5	R5	0.03473	0.08517	0.1041	0.08124	0.6448	0.6078
4/5	R6	0.03387	0.0858	0.1071	0.08278	0.6488	0.6119
4/5	R7	0.03431	0.08791	0.1058	0.08107	0.6486	0.6116
4/5	R8	0.03498	0.08563	0.1069	0.08291	0.6508	0.6138
4/5	R9	0.03437	0.08678	0.1039	0.08424	0.65	0.613
4/5	R10	0.03419	0.08665	0.106	0.08284	0.6527	0.6156
4/5	Myopic	0.004207	0.08421	0.1005	0.07823	0.6022	0.5653
4/5	Base stock	0.001781	0.08706	0.09759	0.08174	0.6061	0.5691
4/5	Two-period	0.06739	0.08129	0.08222	0.07834	0.6406	0.6036
4/5	N32/2/303	3.278	0.08119	0.08017	0.08021	3.856	3.819
4/5	N32/2/304	3.28	0.08277	0.07986	0.08126	3.864	3.827
4/5	N32/2/305	3.3	0.08139	0.08086	0.08053	3.882	3.845
5/4	N32/2/303	4.424	0.1562	0.1362	0.1488	5.439	5.438
5/4	P2	0.009263	0.16	0.1456	0.1579	1.065	1.065
5/4	R1	0.05625	0.1695	0.1838	0.1585	1.176	1.176
5/4	Myopic	0.007529	0.1646	0.1783	0.1504	1.089	1.089
5/4	Base stock	0.002879	0.1646	0.1526	0.1534	1.062	1.062
5/4	Two-period	0.1317	0.1599	0.1471	0.1604	1.18	1.18
6/4	N32/2/303	13.33	0.7603	0.6143	0.7353	18.46	18.45
6/4	P2	0.04243	0.8219	0.6497	0.7728	5.301	5.301
6/4	R1	0.2291	0.8564	0.8053	0.784	5.825	5.825
6/4	Myopic	0.03442	0.8714	0.8977	0.8448	5.867	5.867
6/4	Base stock	0.01109	0.867	0.7107	0.7889	5.48	5.48
6/4	Two-period	0.7142	0.8352	0.6834	0.799	6.15	6.149
7/4	N32/2/303	51.6	4.137	3.356	4.438	79.84	79.83
7/4	P2	0.2275	4.766	3.542	4.659	29.81	29.81
7/4	R1	0.8988	5.056	4.315	4.697	31.92	31.91
7/4	Myopic	0.1622	5.249	4.151	4.677	31	31
7/4	Base stock	0.04446	5.209	3.812	4.946	31.08	31.08
7/4	Two-period	4.105	5.068	3.411	4.531	33.8	33.8

TABLE III: Deficits of raw actions against completed suffixes under the budget rule

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
2/4	N16/1/301	0	0	0	0	80
2/4	N16/1/302	0	0	0	0	80

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
2/4	N16/2/301	0	0	0	0	80
2/4	N16/2/302	0	0	0	0	80
2/4	N32/1/301	0	0	0	0	80
2/4	N32/1/302	0	0	0	0	80
2/4	N32/2/301	0	0	0	0	80
2/4	N32/2/302	0	0	0	0	80
2/4	P2	0	0	0.1119	0.1211	80
2/4	P3	0	0	0.1119	0.1211	80
2/4	R1	0.4343	1.192	2.131	2.662	80
2/4	R2	0.2907	1.439	2.192	2.728	80
2/4	R3	0.2376	1.378	2.871	2.971	80
2/4	R4	0.2114	1.518	2.617	3.148	80
2/4	R5	0.2518	1.301	2.391	2.905	80
2/4	R6	0.1725	1.341	1.996	3.148	80
2/4	R7	0.1633	1.294	2.617	2.905	80
2/4	R8	0.1517	1.274	3.102	3.148	80
2/4	R9	0.168	1.549	2.859	3.148	80
2/4	R10	0.2359	1.378	2.628	2.971	80
2/4	Myopic	0.5401	1.631	2.681	2.971	80
2/4	Base stock	1.045×10^{-4}	0.5148	1.097	1.196	80
2/4	Two-period	0	0.2114	0.4312	0.4433	80
3/4	N16/1/301	0	0	0	0.01418	84
3/4	N16/1/302	0	0	0	0.01418	84
3/4	N16/2/301	0	0	0	0	84
3/4	N16/2/302	0	0	0	0	84
3/4	N32/1/301	0	0	0	0.01355	84
3/4	N32/1/302	0	0	0	0.01418	84
3/4	N32/2/301	0	0	0	0.01355	84
3/4	N32/2/302	0	0	0	0.01418	84
3/4	P2	0	0	0	0	84
3/4	P3	0	0	0	0	84
3/4	R1	0.4902	1.254	1.805	1.997	84
3/4	R2	0.5278	1.396	1.982	2.036	84
3/4	R3	0.5257	1.208	1.952	2.036	84
3/4	R4	0.5869	1.301	1.931	2.056	84
3/4	R5	0.4777	1.16	1.843	2.036	84
3/4	R6	0.5278	1.273	1.807	2.056	84
3/4	R7	0.4736	1.158	1.843	2.017	84
3/4	R8	0.5426	1.271	1.884	2.056	84
3/4	R9	0.4433	1.307	1.843	2.056	84
3/4	R10	0.4778	1.328	1.983	2.017	84
3/4	Myopic	0.7182	1.203	1.57	1.845	84
3/4	Base stock	0	0.5577	0.766	0.8601	84
3/4	Two-period	0	0.326	0.5627	0.62	84
4/4	N16/1/301	0	0	0.1632	0.2253	86
4/4	N16/1/302	0	0	0.1632	0.2253	86
4/4	N16/2/301	0	0	0.1632	0.2253	86
4/4	N16/2/302	0	0	0.1632	0.2253	86
4/4	N32/1/301	0	0	0.1632	0.2253	86
4/4	N32/1/302	0	0	0.1632	0.2253	86
4/4	N32/2/301	0	0	0.1632	0.2253	86
4/4	N32/2/302	0	0	0.1632	0.2253	86
4/4	P2	0	0	0	0.03934	86
4/4	P3	0	0	0.1632	0.2253	86
4/4	R1	0.6107	1.392	1.73	2.306	86
4/4	R2	0.6168	1.371	1.717	1.86	86

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
4/4	R3	0.6243	1.368	1.73	2.306	86
4/4	R4	0.6215	1.378	1.726	2.281	86
4/4	R5	0.6111	1.306	1.716	2.281	86
4/4	R6	0.6111	1.31	1.717	2.281	86
4/4	R7	0.6259	1.379	1.73	2.306	86
4/4	R8	0.5938	1.308	1.728	2.281	86
4/4	R9	0.6215	1.326	1.733	2.306	86
4/4	R10	0.5956	1.318	1.73	2.306	86
4/4	Myopic	0.7088	1.164	1.502	1.611	86
4/4	Base stock	0	0.5887	0.7004	0.7239	86
4/4	Two-period	0	0.4476	0.692	0.6978	86
4/4	N32/2/303	0	0	0.1632	0.2253	86
4/4	N32/2/304	0	0	0.1632	0.2253	86
4/4	N32/2/305	0	0	0.1632	0.2253	86
4/4	N32/2/306	0	0	0.1632	0.2253	86
4/4	N32/2/307	0	0	0.1632	0.2253	86
4/4	N32/2/308	0	0	0.1632	0.2253	86
4/4	N32/2/309	0	0	0.1632	0.2253	86
4/4	N32/2/310	0	0	0.1632	0.2253	86
4/5	N32/2/301	0	0	0.04569	0.225	87
4/5	N32/2/302	0	0	0.04569	0.225	87
4/5	P2	0	0	0.02476	0.1577	87
4/5	P3	0	0	0.006551	0.1029	87
4/5	R1	0.5853	1.342	1.802	2.306	87
4/5	R2	0.592	1.342	1.864	2.306	87
4/5	R3	0.6193	1.347	1.864	2.281	87
4/5	R4	0.5852	1.35	1.848	2.306	87
4/5	R5	0.592	1.342	1.864	2.306	87
4/5	R6	0.6215	1.352	1.864	2.281	87
4/5	R7	0.6215	1.347	1.864	2.281	87
4/5	R8	0.5958	1.351	1.864	2.306	87
4/5	R9	0.592	1.357	1.864	2.306	87
4/5	R10	0.5796	1.321	1.848	2.281	87
4/5	Myopic	0.6812	1.129	1.473	1.611	87
4/5	Base stock	0	0.6134	0.9805	1.143	87
4/5	Two-period	0	0.5771	0.9805	1.143	87
4/5	N32/2/303	0	0	0.03951	0.225	87
4/5	N32/2/304	0	0	0.04426	0.225	87
4/5	N32/2/305	0	0	0.03951	0.225	87
5/4	N32/2/303	0	0	0	0.2398	89
5/4	P2	0	0	0	0.2398	89
5/4	R1	0.7081	1.468	1.816	2.547	89
5/4	Myopic	0.7604	1.014	1.474	1.856	89
5/4	Base stock	0	0.5978	0.7643	0.8984	89
5/4	Two-period	0	0.5978	0.7643	1.16	89
6/4	N32/2/303	0	0	0	0.7549	91
6/4	P2	0	0	0.0178	0.7549	91
6/4	R1	0.7964	1.544	2.24	3.164	91
6/4	Myopic	0.833	0.9146	1.542	2.089	91
6/4	Base stock	0	0.531	0.8061	1.425	91
6/4	Two-period	0	0.5293	0.8061	2.217	91
7/4	N32/2/303	0	0	0	1.23	93
7/4	P2	0	0	0.5536	1.23	93
7/4	R1	0.8443	1.596	2.685	4.162	93
7/4	Myopic	0.8503	0.9031	1.557	2.315	93
7/4	Base stock	0	0.4119	0.8073	1.926	93

d/L	Policy	Median	90th pct.	99th pct.	Maximum	Bits
7/4	Two-period	0	0.3865	0.8125	3.235	93

TABLE IV: Corrections by time and inventory boundary; worst losses over all candidates in a model

d/L	Time	Edits	Min inventory 0	Raw restart loss	Budget restart loss
2/4	0	121	56	5.054	4.78×10^{-5}
2/4	1	135	57	4.485	5.699×10^{-5}
2/4	2	125	58	4.134	6.396×10^{-5}
2/4	3	123	57	3.905	1.034×10^{-4}
2/4	4	125	53	3.434	2.089×10^{-4}
2/4	5	123	48	3.042	0
2/4	6	115	51	2.296	0
2/4	7	104	49	1.677	0
3/4	0	531	305	5.141	1.065×10^{-4}
3/4	1	542	311	4.96	1.416×10^{-4}
3/4	2	540	306	4.776	2.009×10^{-4}
3/4	3	538	311	4.732	2.841×10^{-4}
3/4	4	541	315	3.937	4.099×10^{-4}
3/4	5	522	307	3.726	0.001035
3/4	6	516	306	3.013	0.001307
3/4	7	492	292	2.017	0.006601
4/4	0	2478	1732	5.736	0.002999
4/4	1	2531	1805	5.539	7.755×10^{-5}
4/4	2	2483	1763	5.285	2.566×10^{-4}
4/4	3	2241	1507	5.194	3.834×10^{-4}
4/4	4	2231	1514	4.516	0.001808
4/4	5	2272	1515	3.918	0.006259
4/4	6	2099	1513	3.323	0.009483
4/4	7	2197	1510	2.306	0
4/5	0	6068	3465	6.462	2.001×10^{-4}
4/5	1	5977	3379	6.933	2.327×10^{-4}
4/5	2	5936	3381	5.975	9.397×10^{-4}
4/5	3	5892	3308	6.299	0.001218
4/5	4	5883	3337	5.165	0.006349
4/5	5	5920	3354	4.331	0.005074
4/5	6	5455	3303	3.514	0.009483
4/5	7	5561	3324	2.306	0
5/4	0	2305	1422	7.066	9.741×10^{-6}
5/4	1	2290	1431	6.407	3.449×10^{-5}
5/4	2	2265	1411	5.962	9.435×10^{-5}
5/4	3	2275	1412	5.28	1.572×10^{-4}
5/4	4	2271	1411	4.638	3.627×10^{-4}
5/4	5	2245	1394	4.114	0.001283
5/4	6	1798	1381	3.076	0.009065
5/4	7	1804	1399	2.547	0
6/4	0	8874	6144	7.541	7.463×10^{-8}
6/4	1	8812	6087	7.363	5.196×10^{-7}
6/4	2	8781	6070	6.573	2.43×10^{-6}
6/4	3	8840	6103	5.786	9.088×10^{-6}
6/4	4	8743	6117	5.144	2.155×10^{-5}
6/4	5	8745	6126	4.346	4.748×10^{-5}
6/4	6	7343	6028	3.392	1.279×10^{-4}

d/L	Time	Edits	Min inventory 0	Raw restart loss	Budget restart loss
6/4	7	7285	6015	2.796	5.167×10^{-4}
7/4	0	34200	26094	10.41	3.438×10^{-7}
7/4	1	34230	26124	9.569	3.126×10^{-6}
7/4	2	33712	25577	8.717	2.558×10^{-5}
7/4	3	33751	25589	7.72	1.938×10^{-4}
7/4	4	33816	25661	6.182	0.001292
7/4	5	33674	25733	4.863	0.006405
7/4	6	29333	25486	3.927	0.009704
7/4	7	29418	25487	3.032	0

3. INHERITED R28 DETAILED EXECUTION LEDGERS

These are the original R28 hardware measurements, not cross-hardware speedup comparisons. Their underlying records remain byte-for-byte unchanged.

TABLE V: All original R28 inventory candidates

d, L	Policy	Raw bound	Raw loss	Complete bound	Complete loss	Edits
2,4	N16/1/301	0	0	0	0	0
2,4	N16/1/302	0	0	0	0	0
2,4	N16/2/301	0	0	0	0	0
2,4	N16/2/302	0	0	0	0	0
2,4	N32/1/301	0	0	0	0	0
2,4	N32/1/302	0	0	0	0	0
2,4	N32/2/301	0	0	0	0	0
2,4	N32/2/302	0	0	0	0	0
2,4	P2	0.1571	0.1391	2.089×10^{-4}	2.089×10^{-4}	7
2,4	P3	0.1571	0.1391	0	0	7
3,4	N16/1/301	0.01418	0.01418	0	0	3
3,4	N16/1/302	0.01418	0.01418	0	0	3
3,4	N16/2/301	0	0	0	0	0
3,4	N16/2/302	0	0	0	0	0
3,4	N32/1/301	0.01355	0.01355	0	0	2
3,4	N32/1/302	0.01418	0.01418	0	0	1
3,4	N32/2/301	0.01355	0.01355	0	0	1
3,4	N32/2/302	0.01418	0.01418	0	0	3
3,4	P2	0	0	0	0	0
3,4	P3	0	0	0	0	0
4,4	N16/1/301	0.4915	0.3666	0	0	61
4,4	N16/1/302	0.4914	0.3666	0	0	59
4,4	N16/2/301	0.4915	0.3666	0	0	55
4,4	N16/2/302	0.4915	0.3666	0	0	56
4,4	N32/1/301	0.4914	0.3666	0	0	61
4,4	N32/1/302	0.4914	0.3666	0	0	59
4,4	N32/2/301	0.4915	0.3666	0	0	61
4,4	N32/2/302	0.4915	0.3666	0	0	58
4,4	P2	0.03934	0.03934	0	0	17
4,4	P3	0.4914	0.3666	0	0	59
4,5	N32/2/301	0.4913	0.3664	3.962×10^{-4}	3.962×10^{-4}	112
4,5	N32/2/302	0.4913	0.3664	3.962×10^{-4}	3.962×10^{-4}	120
4,5	P2	0.2019	0.1925	3.962×10^{-4}	3.962×10^{-4}	92
4,5	P3	0.1108	0.1029	3.962×10^{-4}	3.962×10^{-4}	84

TABLE VI: Original R28 work and correction ledger

d, L	Policy	Build	Compile	Ref.-free	Standalone	% edits	99th loss
2,4	N16/1/301	6.704×10^{-4}	6.738×10^{-4}	1.505	1.508	0	0
2,4	N16/1/302	4.819×10^{-4}	5.66×10^{-4}	1.497	1.501	0	0
2,4	N16/2/301	4.684×10^{-4}	5.76×10^{-4}	1.862	1.865	0	0
2,4	N16/2/302	4.557×10^{-4}	6.148×10^{-4}	1.863	1.866	0	0
2,4	N32/1/301	4.304×10^{-4}	5.766×10^{-4}	1.526	1.529	0	0
2,4	N32/1/302	4.384×10^{-4}	5.761×10^{-4}	1.53	1.534	0	0
2,4	N32/2/301	4.63×10^{-4}	5.88×10^{-4}	1.982	1.985	0	0
2,4	N32/2/302	4.552×10^{-4}	6.102×10^{-4}	2.092	2.095	0	0
2,4	P2	4.006×10^{-4}	2.652×10^{-4}	0.007922	0.01115	5.833	0.1212
2,4	P3	3.829×10^{-4}	2.462×10^{-4}	0.007613	0.0108	5.833	0.1212
3,4	N16/1/301	0.001102	8.04×10^{-4}	1.555	1.569	0.5952	0.005613
3,4	N16/1/302	0.001103	7.941×10^{-4}	1.567	1.581	0.5952	0.002806
3,4	N16/2/301	0.001148	8.265×10^{-4}	1.977	1.991	0	0
3,4	N16/2/302	0.001173	7.869×10^{-4}	1.989	2.003	0	0
3,4	N32/1/301	0.001166	8.163×10^{-4}	1.628	1.642	0.3968	1.051×10^{-4}
3,4	N32/1/302	0.001148	7.93×10^{-4}	1.63	1.644	0.1984	0.002806
3,4	N32/2/301	0.001194	8.188×10^{-4}	2.174	2.188	0.1984	0
3,4	N32/2/302	0.001213	8.081×10^{-4}	2.175	2.189	0.5952	0.008419
3,4	P2	0.001105	4.383×10^{-4}	0.02348	0.03774	0	0
3,4	P3	0.001157	4.629×10^{-4}	0.02406	0.03833	0	0
4,4	N16/1/301	0.006404	0.001782	1.78	1.855	2.99	0.2052
4,4	N16/1/302	0.005622	0.001828	1.782	1.857	2.892	0.2052
4,4	N16/2/301	0.005445	0.00179	2.368	2.442	2.696	0.2052
4,4	N16/2/302	0.005581	0.00176	2.369	2.444	2.745	0.2052
4,4	N32/1/301	0.005494	0.001831	1.958	2.033	2.99	0.2052
4,4	N32/1/302	0.005471	0.001807	1.95	2.025	2.892	0.2052
4,4	N32/2/301	0.005458	0.001779	2.974	3.048	2.99	0.2052
4,4	N32/2/302	0.005549	0.001773	2.963	3.038	2.843	0.2052
4,4	P2	0.005477	0.00141	0.1017	0.1762	0.8333	0.00649
4,4	P3	0.005537	0.001415	0.1036	0.1782	2.892	0.2052
4,5	N32/2/301	0.02482	0.003514	4.392	4.586	2.244	0.1706
4,5	N32/2/302	0.01684	0.003527	4.386	4.578	2.404	0.1706
4,5	P2	0.01272	0.003051	0.2448	0.4368	1.843	0.05718
4,5	P3	0.01273	0.003075	0.2451	0.4385	1.683	0.01822

TABLE VII: Original R28 restart losses

d, L	Time	Raw worst	Complete worst
2,4	0	0.01797	4.78×10^{-5}
2,4	1	0.019	5.699×10^{-5}
2,4	2	0.02281	6.396×10^{-5}
2,4	3	0.02997	1.034×10^{-4}
2,4	4	0.03888	2.089×10^{-4}
2,4	5	0.1391	0
2,4	6	0.0728	0
2,4	7	0	0
3,4	0	5.014×10^{-4}	0
3,4	1	6.384×10^{-4}	0
3,4	2	6.754×10^{-4}	0
3,4	3	0.001651	0

d, L	Time	Raw worst	Complete worst
3,4	4	0.002779	0
3,4	5	0.00842	0
3,4	6	0.01418	0
3,4	7	0	0
4,4	0	0.3666	0
4,4	1	0.2068	0
4,4	2	0.04572	0
4,4	3	0.001242	0
4,4	4	0.007523	0
4,4	5	0.0456	0
4,4	6	0.0325	0
4,4	7	0	0
4,5	0	0.3664	1.856×10^{-4}
4,5	1	0.2067	6.536×10^{-5}
4,5	2	0.04569	3.962×10^{-4}
4,5	3	0.08895	0
4,5	4	0.1925	0
4,5	5	0.05719	0
4,5	6	0.1156	0
4,5	7	0	0

TABLE VIII: Recentered transport timing

Seed	Order	Setup	Neural	Transport	Diagnostics	Total
27101	8/16	0.003616	0.5799	0.8522	0.06444	36.62
27101	12/24	0.002385	0.8794	1.157	0.06613	37.18
27102	8/16	0.002306	0.578	0.8489	0.06433	1.507
27102	12/24	0.002181	0.8853	1.165	0.06696	2.135
27103	8/16	0.002029	0.5797	0.851	0.06446	1.511
27103	12/24	0.002136	0.8814	1.158	0.06652	2.124

TABLE IX: Paired production timing

Seed	Method	Setup	Generation	Checker	Bookkeeping	Total
27201	Neural Adam U	0.001439	1.174	87.31	0.00562	88.54
27201	Neural Adam G	0.00157	1.173	87.1	0.005491	88.33
27201	Direct Adam U	0.001669	1.047	87.43	0.00343	88.53
27201	Direct Adam G	0.00168	1.057	87.58	0.003551	88.68
27201	Direct L-BFGS-B U	0.001572	0.298	87.34	0.003031	87.69
27201	Direct L-BFGS-B G	0.001619	0.354	87.31	0.005319	87.71
27202	Neural Adam U	0.001469	1.163	87.3	0.005566	88.52
27202	Neural Adam G	0.00149	1.163	87.6	0.005498	88.82
27202	Direct Adam U	0.001584	1.045	87.35	0.003501	88.44
27202	Direct Adam G	0.001564	1.047	87.22	0.003501	88.31
27202	Direct L-BFGS-B U	0.00157	0.2793	87.5	0.002768	87.82
27202	Direct L-BFGS-B G	0.001563	0.3027	87.29	0.005414	87.64

TABLE X: Every production checkpoint decision

Seed	Method	Block	Lower	Upper	Margin	Width	Decision
27201	Neural Adam U	1	-1.28723	-1.287229	0.03313	3.938×10^{-7}	D
27201	Neural Adam U	2	-1.287225	-1.287224	4.549×10^{-6}	3.938×10^{-7}	D
27201	Neural Adam U	3	-1.287223	-1.287222	1.913×10^{-6}	3.938×10^{-7}	D
27201	Neural Adam U	4	-1.287222	-1.287221	6.623×10^{-7}	3.938×10^{-7}	D
27201	Neural Adam G	1	-1.28723	-1.287229	0.03313	3.938×10^{-7}	A
27201	Neural Adam G	2	-1.287225	-1.287224	4.549×10^{-6}	3.938×10^{-7}	A
27201	Neural Adam G	3	-1.287223	-1.287222	1.913×10^{-6}	3.938×10^{-7}	A
27201	Neural Adam G	4	-1.287222	-1.287221	6.623×10^{-7}	3.938×10^{-7}	A
27201	Direct Adam U	1	-1.287282	-1.28728	0.03308	3.938×10^{-7}	D
27201	Direct Adam U	2	-1.287251	-1.287249	3.056×10^{-5}	3.938×10^{-7}	D
27201	Direct Adam U	3	-1.28724	-1.287238	1.082×10^{-5}	3.938×10^{-7}	D
27201	Direct Adam U	4	-1.287234	-1.287232	5.481×10^{-6}	3.938×10^{-7}	D
27201	Direct Adam G	1	-1.287282	-1.28728	0.03308	3.938×10^{-7}	A
27201	Direct Adam G	2	-1.287251	-1.287249	3.056×10^{-5}	3.938×10^{-7}	A
27201	Direct Adam G	3	-1.28724	-1.287238	1.082×10^{-5}	3.938×10^{-7}	A
27201	Direct Adam G	4	-1.287234	-1.287232	5.481×10^{-6}	3.938×10^{-7}	A
27201	Direct L-BFGS-B U	1	-1.28722	-1.287219	0.03314	3.938×10^{-7}	D
27201	Direct L-BFGS-B U	2	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	D
27201	Direct L-BFGS-B U	3	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27201	Direct L-BFGS-B U	4	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27201	Direct L-BFGS-B G	1	-1.28722	-1.287219	0.03314	3.938×10^{-7}	A
27201	Direct L-BFGS-B G	2	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	R
27201	Direct L-BFGS-B G	3	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	R
27201	Direct L-BFGS-B G	4	-1.28722	-1.287219	-3.936×10^{-7}	3.938×10^{-7}	R
27202	Neural Adam U	1	-1.28723	-1.287229	0.03274	3.938×10^{-7}	D
27202	Neural Adam U	2	-1.287225	-1.287224	4.779×10^{-6}	3.938×10^{-7}	D
27202	Neural Adam U	3	-1.287223	-1.287222	1.253×10^{-6}	3.938×10^{-7}	D
27202	Neural Adam U	4	-1.287222	-1.287221	5.336×10^{-7}	3.938×10^{-7}	D
27202	Neural Adam G	1	-1.28723	-1.287229	0.03274	3.938×10^{-7}	A
27202	Neural Adam G	2	-1.287225	-1.287224	4.779×10^{-6}	3.938×10^{-7}	A
27202	Neural Adam G	3	-1.287223	-1.287222	1.253×10^{-6}	3.938×10^{-7}	A
27202	Neural Adam G	4	-1.287222	-1.287221	5.336×10^{-7}	3.938×10^{-7}	A
27202	Direct Adam U	1	-1.287282	-1.28728	0.03269	3.938×10^{-7}	D
27202	Direct Adam U	2	-1.287251	-1.287249	3.057×10^{-5}	3.938×10^{-7}	D
27202	Direct Adam U	3	-1.287239	-1.287238	1.078×10^{-5}	3.938×10^{-7}	D
27202	Direct Adam U	4	-1.287234	-1.287232	5.455×10^{-6}	3.938×10^{-7}	D
27202	Direct Adam G	1	-1.287282	-1.28728	0.03269	3.938×10^{-7}	A
27202	Direct Adam G	2	-1.287251	-1.287249	3.057×10^{-5}	3.938×10^{-7}	A
27202	Direct Adam G	3	-1.287239	-1.287238	1.078×10^{-5}	3.938×10^{-7}	A
27202	Direct Adam G	4	-1.287234	-1.287232	5.455×10^{-6}	3.938×10^{-7}	A
27202	Direct L-BFGS-B U	1	-1.28722	-1.287219	0.03275	3.938×10^{-7}	D
27202	Direct L-BFGS-B U	2	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	D
27202	Direct L-BFGS-B U	3	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27202	Direct L-BFGS-B U	4	-1.28722	-1.287219	-3.938×10^{-7}	3.938×10^{-7}	D
27202	Direct L-BFGS-B G	1	-1.28722	-1.287219	0.03275	3.938×10^{-7}	A
27202	Direct L-BFGS-B G	2	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	R
27202	Direct L-BFGS-B G	3	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	R
27202	Direct L-BFGS-B G	4	-1.28722	-1.287219	-3.939×10^{-7}	3.938×10^{-7}	R